

API-INTEGRATION STRATEGIE

Domain Forensic Analyzer - SOC-Analyst-Level Intelligence

ZIELSETZUNG: SOC-ANALYST-AUTOMATION

REAL-WORLD SCENARIO:

SOC Alert: Suspicious domain "malware-example.com" detected in network traffic

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Analyst Action: Manual threat intelligence gathering (15-20 minutes)

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Automated Solution: Domain Forensic Analyzer (2-3 minutes)

SOC-ANALYST-WORKFLOW AUTOMATION:

1. **Domain Reputation Check** → VirusTotal
2. **Infrastructure Analysis** → SecurityTrails Historical Data
3. **IP Reputation Assessment** → AbuseIPDB
4. **Risk Correlation** → Multi-source intelligence fusion
5. **Actionable Intelligence** → Immediate threat classification

API-INTEGRATION ARCHITEKTUR

MULTI-SOURCE THREAT INTELLIGENCE FUSION

LAYER 1: INFRASTRUCTURE INTELLIGENCE

SecurityTrails API

- Historical DNS Records → Infrastructure Timeline
- Subdomain Discovery → Attack Surface Mapping
- DNS Changes → Suspicious Migrations
- Certificate History → SSL/TLS Intelligence

LAYER 2: REPUTATION INTELLIGENCE

VirusTotal API

- Domain Classification → Malware/Phishing Detection
- URL Analysis → Threat Categories
- Malicious Score → Risk Quantification
- Community Reports → Analyst Insights

LAYER 3: BEHAVIORAL INTELLIGENCE

AbuseIPDB API

- IP Reputation Scoring → Geographic Risk
- Attack History → Behavioral Patterns

THREAT INTELLIGENCE FUSION FRAMEWORK

API RESPONSE PROCESSING

1. SECURITYTRAILS PROCESSING

```
def process_securitytrails_intelligence(api_response):  
    """  
    Extract forensic intelligence from SecurityTrails  
    """  
    intelligence = {  
        'historical_subdomains': api_response.get('subdomains', []),  
        'dns_timeline': [],  
        'infrastructure_changes': [],  
        'suspicious_patterns': []  
    }  
  
    # DNS History Analysis  
    for record in api_response.get('dns_history', []):  
        intelligence['dns_timeline'].append({  
            'date': record['first_seen'],  
            'ip': record['values'][0]['ip'],  
            'change_type': 'infrastructure_migration'  
        })  
  
    # Pattern Detection  
    if len(intelligence['dns_timeline']) > 5:  
        intelligence['suspicious_patterns'].append('frequent_dns_changes')  
  
    return intelligence
```

2. VIRUSTOTAL PROCESSING

```
def process_virustotal_intelligence(api_response):  
    """  
    Extract threat intelligence from VirusTotal  
    """  
    stats = api_response['data']['attributes']['last_analysis_stats']  
  
    threat_intelligence = {  
        'malicious_detections': stats.get('malicious', 0),  
        'suspicious_detections': stats.get('suspicious', 0),  
        'total_engines': sum(stats.values()),  
        'threat_categories': api_response['data']['attributes'].get('categories', []),  
        'reputation_score': api_response['data']['attributes'].get('reputation', 0)  
    }  
  
    # Risk Classification  
    if threat_intelligence['malicious_detections'] > 0:  
        threat_intelligence['risk_level'] = 'CRITICAL'
```

```

elif threat_intelligence['suspicious_detections'] > 2:
    threat_intelligence['risk_level'] = 'HIGH'
else:
    threat_intelligence['risk_level'] = 'CLEAN'

return threat_intelligence

```

3. ABUSEIPDB PROCESSING

```

def process_abuseipdb_intelligence(api_response):
    """
    Extract IP reputation intelligence from AbuseIPDB
    """
    reputation_intelligence = {
        'abuse_confidence': api_response.get('abuseConfidencePercentage', 0),
        'country_code': api_response.get('countryCode', 'Unknown'),
        'is_whitelisted': api_response.get('isWhitelisted', False),
        'recent_reports': len(api_response.get('reports', [])),
        'attack_categories': []
    }

    # Attack Pattern Analysis
    for report in api_response.get('reports', [])[:5]: # Last 5 reports
        categories = report.get('categories', [])
        reputation_intelligence['attack_categories'].extend(categories)

    # Risk Assessment
    if reputation_intelligence['abuse_confidence'] > 75:
        reputation_intelligence['ip_risk'] = 'HIGH'
    elif reputation_intelligence['abuse_confidence'] > 25:
        reputation_intelligence['ip_risk'] = 'MEDIUM'
    else:
        reputation_intelligence['ip_risk'] = 'LOW'

    return reputation_intelligence

```

ENHANCED RISK ASSESSMENT ALGORITHM

MULTI-SOURCE RISK CORRELATION

```

def calculate_enhanced_risk_score(domain_analysis_result):
    """
    SOC-Analyst-Level Risk Assessment using Multi-API Intelligence
    """
    base_score = domain_analysis_result.risk_score # From existing analysis

    # VirusTotal Impact
    vt_intelligence = domain_analysis_result.virustotal_intelligence
    if vt_intelligence['risk_level'] == 'CRITICAL':
        base_score += 8.0
    elif vt_intelligence['risk_level'] == 'HIGH':
        base_score += 5.0

```

```

# AbuseIPDB Impact
abuse_intelligence = domain_analysis_result.abuseipdb_intelligence
if abuse_intelligence['ip_risk'] == 'HIGH':
    base_score += 3.0
elif abuse_intelligence['ip_risk'] == 'MEDIUM':
    base_score += 1.5

# SecurityTrails Patterns
st_intelligence = domain_analysis_result.securitytrails_intelligence
if 'frequent_dns_changes' in st_intelligence['suspicious_patterns']:
    base_score += 2.0

# Final Score Normalization (0.0 - 10.0)
final_score = min(10.0, base_score)

return {
    'enhanced_risk_score': final_score,
    'threat_classification': classify_threat_level(final_score),
    'actionable_intelligence': generate_analyst_recommendations(domain_analy
}

def classify_threat_level(score):
    """SOC-Standard Threat Classification"""
    if score >= 8.0:
        return 'CRITICAL - Immediate Action Required'
    elif score >= 6.0:
        return 'HIGH - Priority Investigation'
    elif score >= 4.0:
        return 'MEDIUM - Monitoring Recommended'
    elif score >= 2.0:
        return 'LOW - Informational'
    else:
        return 'CLEAN - No Action Required'

```

SOC-ANALYST ACTIONABLE INTELLIGENCE

AUTOMATED ANALYST RECOMMENDATIONS

```

def generate_analyst_recommendations(analysis_result):
    """
    Generate SOC-Analyst-Ready Action Items
    """
    recommendations = []

    # VirusTotal-Based Actions
    vt = analysis_result.virustotal_intelligence
    if vt['malicious_detections'] > 0:
        recommendations.append({
            'priority': 'CRITICAL',
            'action': 'IMMEDIATE_BLOCK',
            'reason': f"Domain flagged by {vt['malicious_detections']} security

```

```

        'evidence': vt['threat_categories']
    })

# AbuseIPDB-Based Actions
abuse = analysis_result.abuseipdb_intelligence
if abuse['abuse_confidence'] > 50:
    recommendations.append({
        'priority': 'HIGH',
        'action': 'IP_INVESTIGATION',
        'reason': f"High abuse confidence: {abuse['abuse_confidence']}%",
        'evidence': f"Recent attacks from {abuse['country_code']}"
    })

# SecurityTrails-Based Actions
st = analysis_result.securitytrails_intelligence
if len(st['dns_timeline']) > 0:
    recommendations.append({
        'priority': 'MEDIUM',
        'action': 'INFRASTRUCTURE_ANALYSIS',
        'reason': 'Historical DNS changes detected',
        'evidence': f"{len(st['dns_timeline'])} infrastructure migrations"
    })

return recommendations

```

IMPLEMENTATION ROADMAP

PHASE 1: SECURE API INTEGRATION

1. API-Key Management System
 - └ config/api_keys.json (gitignore protected)
 - └ Environment variable loading
 - └ Secure key rotation support
2. Error Handling & Rate Limiting
 - └ API timeout management
 - └ Rate limit respect
 - └ Fallback mechanisms

PHASE 2: INTELLIGENCE MODULES

3. SecurityTrails Enhancement (Demo → Live)
 - └ Historical DNS analysis
 - └ Subdomain timeline mapping
 - └ Infrastructure change detection
4. VirusTotal Integration
 - └ Domain reputation checking
 - └ URL analysis capability
 - └ Threat category classification
5. AbuseIPDB Integration

- └ IP reputation scoring
- └ Geographic risk assessment
- └ Attack pattern analysis

PHASE 3: SOC-LEVEL INTELLIGENCE FUSION

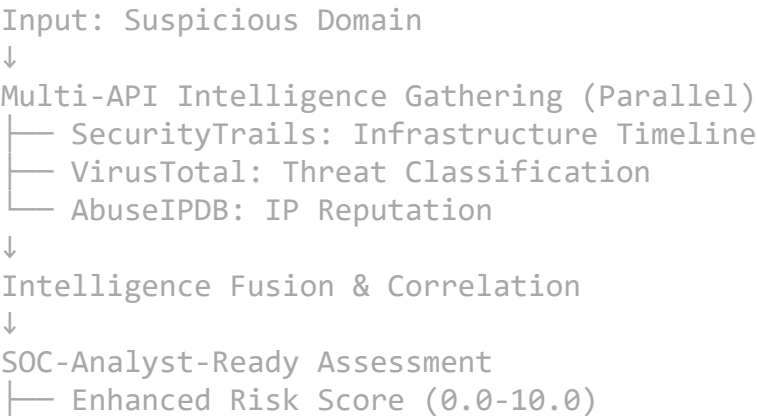
- 6. Enhanced Risk Assessment
 - └ Multi-source correlation
 - └ SOC-standard threat classification
 - └ Automated analyst recommendations
- 7. Professional Reporting
 - └ SOC-analyst-ready summaries
 - └ Actionable intelligence extraction
 - └ Evidence-based recommendations

FORENSIC VALUE MATRIX

INTELLIGENCE CAPABILITY ENHANCEMENT

Current Capability	API Enhancement	SOC Value
Subdomain Discovery	SecurityTrails Historical	Attack Surface Evolution
Basic Risk Assessment	VirusTotal Reputation	Threat Classification
Network Analysis	AbuseIPDB IP Intel	Geographic Threat Context
Manual Investigation	Automated Intelligence	15min → 3min Analysis

PROFESSIONAL THREAT INTELLIGENCE PIPELINE



- Threat Classification
- Actionable Recommendations
- Evidence-Based Analysis



Output: Professional Threat Intelligence Report

SUCCESS METRICS

SOC-ANALYST WORKFLOW IMPROVEMENT

- **Time Reduction:** 15-20 minutes → 2-3 minutes
- **Intelligence Quality:** Multi-source correlation vs single-point analysis
- **Actionable Output:** Automated recommendations vs manual research
- **Professional Standards:** SOC-grade threat classification

PORTFOLIO DEMONSTRATION VALUE

- **Multi-API Integration:** Professional service orchestration
- **Threat Intelligence Fusion:** Advanced correlation techniques
- **Security Analysis:** Real-world SOC analyst capabilities
- **Industry Standards:** Professional threat classification systems